Do Now · retrieval

Lesson 089 · The Processor City: Architecture Overview

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is the function of a program counter?

Q2

What does the ALU perform?

Q3

Name one type of bus.

Q4

A D flip-flop has D=1 and Q=0 just before the active rising edge. What is Q after the edge?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — Inside CPU City Hall, the Control Unit organises operations, the ALU carries out arithmetic and logic, and registers hold immediate working state.

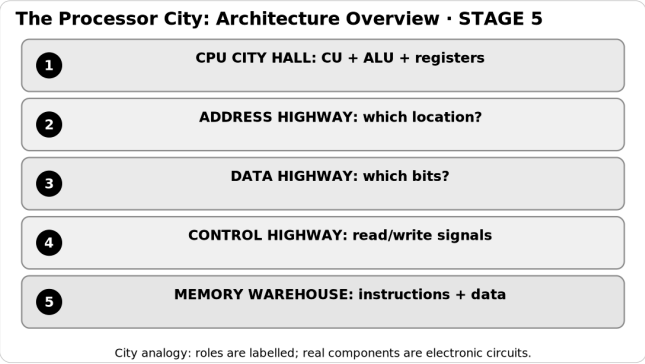
Core idea 2

Explain — The Memory Warehouse stores program instructions and data. Address, data and control buses connect processor and memory; they carry different kinds of information.

Core idea 3

Evaluate — Von Neumann architecture stores instructions and data in a shared memory/address space, whereas Harvard uses separate instruction and data memories/pathways.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

ALU — Arithmetic Logic Unit.

CU — Control Unit.

register — Small fast CPU state storage.

bus — Shared communication pathway.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

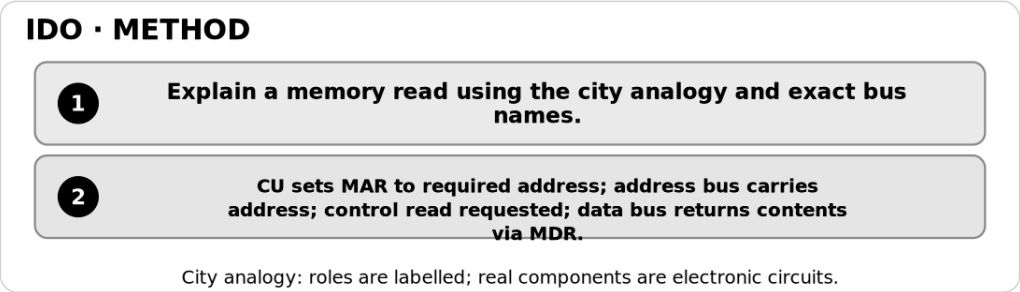
Problem

Explain a memory read using the city analogy and exact bus names.

Follow the worked method

1. Model response — The address identifies where; data is what is transferred. CU coordinates the process.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Model response
Saying the CU performs arithmetic confuses separate roles.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Name and describe ALU, Control Unit, and registers in processor. [6]

Question 2

Using correct names, explain address, data and control bus during memory read. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate usefulness and limitation of Processor City analogy for Von Neumann architecture. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

The data bus carries the address of a memory location.

Use key vocabulary from memory

1. Explain ALU in your own words.

2. Explain CU in your own words.

3. Explain register in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.